

Master Thesis Supervisor Meeting – Authoritative Numbers Cheat Sheet

Mode-I Phase-Field Benchmark Reproduction · Date: 17 September 2026

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1. Authoritative Mechanical Benchmark Numbers

Simulation Model	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]	$u(F_{\max})$ [mm]	Status & Extraction Provenance
Fixed Reference Anchor (1398090.mmmaster02)	15,192 (CPE4)	137.945520 (Ref Anchor)	0.757778 (Ref Anchor)	0.005857 (Ref Anchor)	Unconstrained OLS, $N = 400$ ($u \leq 1.0\,\mu\text{m}$), $R^2 = 0.99999960$, $b = 4.472 \times 10^{-5}$ kN
DEFECTIVE PREPROC. (1399632.mmmaster02) (<i>DIAGNOSTIC ONLY</i>)	71,320	122.379 (−11.28%)	0.478203 (−36.89%)	0.004150 (−29.15%)	DIAGNOSTIC ONLY: 16/150 boundary nodes parsed (boundary set cardinality); bottom lift up to 48.34% stroke
Corrected Frozen 71k (1405044.mmmaster02)	71,320	138.021013 (+0.0547%)	– (frozen elastic)	– (frozen elastic)	Exact frozen requalification, $N = 400$, 150/150 nodes, 0.000 nm lift, 0 warnings
Corrected Nominal 1% (1404933.mmmaster02)	71,320 (69kQ + 1.8kT)	137.820804 (−0.0904%)	0.745325 (−1.64%)	0.005750 (−1.83%)	Authoritative full-fracture solve: $K_0 \approx 137.821$, $N = 400$, $R^2 = 0.99999960$, $b = 4.470 \times 10^{-5}$ kN
Supporting 4T Twin (1405003.mmmaster02)	71,320	137.820804 (−0.0904%)	0.745325 (−1.64%)	0.005750 (−1.83%)	Shared-memory 4-thread parity twin (1 MPI × 4T); tight numerical parity with 1-CPU anchor

2. Canonical Pre-Analysis & Gate-5 Discrepancy Decomposition

A. Canonical Pre-Analysis Baseline (Job PK_M1_PRE):

- Global coarse element seed: $h_{\text{cms}} = 0.020$ mm.
- Total finite elements: 2,906 (2,818 CPE4 + 88 CPE3).
- Total nodes: 2,988 (with duplicate crack seam nodes).
- Extracted MISESeri field: 2,906 **scalar values** at `WHOLE_ELEMENT`.
- Retirement: historical 3,930-row set was Mode-II (1379893); retired.

B. Abaqus Multi-Release Invariance Audit:

- Abaqus 2019 GA (Build 157541): 71,320 **elements**
- Abaqus 2021.HF26 (Job 1404968): 71,320 **elements**
- Abaqus 2022 GA (Job 1404373): 71,320 **elements**
- Abaqus 2023.HF4 (Job 1404373): 71,320 **elements**
- All decks (142,268 lines) are 100.000% **bitwise identical**.

C. Regional Element Distribution (71,320 Reconstruction):

- R1 Crack Tip** ($r \leq 0.02$ mm): 347 elements (0.5%)
- R2 Ligament Corridor** ($y \in [0.48, 0.52]$): 5,318 elements (7.5%)
- R3 Slit Flank Corridor** ($x \leq 0.5$): 3,396 elements (4.8%)
- Process Zone Total (R1+R2+R3)**: 9,061 **elements** (12.7%)
- R4 Transition Zone** ($y \in [0.4, 0.6]$): 20,273 elements (28.4%)
- R5 Far Field** ($y \notin [0.4, 0.6]$): 41,986 elements (58.9%)
- Outside Corridor Total (R4+R5)**: 62,259 **elements** (87.3%)

D. Literature Target vs Reconstruction:

- Literature reported mesh: $\approx 13,941$ **elements**
- Publication-literal 1% reconstruction: 71,320 **elements**
- 2% error target sensitivity: 15,396 **elements** (+10.4%)
- 5% error target sensitivity: 4,194 **elements** (−70.0%)

3. Exhaustive 15-Factor OFAT Audit – Key Numerical Counts

Investigated Factor	Resulting Elements	Shift vs 71,320	Scientific Qualification & Epistemic Status
Top Lateral BC Free ($u_1 = \text{Free}$)	55,761 (Job 1405055)	−21.8%	Reduces far-field stress triaxiality; remains $> 4.0\times$ above literature.
Coarse Seed ($h_{\text{cms}} = 0.030$ mm)	48,919	−31.4%	Coarser donor deck reduces total adapted nodes; remains $> 3.5\times$ above lit.
Plane Stress Formulation (CPS4)	58,679	−17.7%	Confounded: violates standard Mode-I plane strain constraint ($\nu = 0.3$).
Reduced Integration (CPE4R)	103,706 (Job 1405056)	+45.4%	Hourglass modes increase recovery error; drives count further from literature.
Coarsening Dynamics Enabled	71,037	−0.4%	Published rule sets <code>coarseningFactor=NOT_ALLOWED</code> ; minimal effect.
Sizing Bounds Removed	71,320	0.0%	Bound limits ($h_{\min} = 0.001$, $h_{\max} = 0.020$) do not truncate sizing.